WEEK 4 REPORT
Analysis of YOLOv6 Dataset Configuration Files
Introduction
YOLOv6 uses YAML configuration files to define dataset metadata, including image paths, annotations, class names, and dataset properties. These files enable the model to locate training and validation data during object detection training.

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1. COCO Dataset Configuration Analysis (coco.yaml)
The COCO dataset configuration contains:

train: ../coco/images/train2017
val: ../coco/images/val2017
test: ../coco/images/test2017

anno_path: ../coco/annotations/instances_val2017.json

nc: 80
is_coco: True

Observations
- Training images are stored in train2017.
- Validation images are stored in val2017.
- Test images are stored in test2017.
- Annotation information is stored in a JSON file.
- nc: 80 indicates 80 object classes.
- is_coco: True specifies that the dataset follows COCO format.

Sample Classes
The dataset includes:
- Person
- Bicycle
- Car
- Motorcycle
- Airplane
- Bus
- Train
- Truck
- Boat
- Traffic Light

and a total of 80 classes.

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2. VOC Dataset Configuration Analysis (voc.yaml)
The VOC configuration contains:

train: VOCdevkit/voc_07_12/images/train

val: VOCdevkit/voc_07_12/images/val
test: VOCdevkit/voc_07_12/images/val

is_coco: False
nc: 20

Observations

- Uses Pascal VOC dataset.
- Contains 20 object categories.
- Training and validation images are stored separately.
- is_coco: False indicates non-COCO dataset format.

VOC Classes

- Aeroplane
- Bicycle
- Bird
- Boat
- Bottle
- Bus
- Car
- Cat
- Chair
- Cow
- Dining Table
- Dog
- Horse
- Motorbike
- Person
- Potted Plant
- Sheep
- Sofa
- Train
- TV Monitor

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3. Custom Dataset Configuration Analysis (dataset.yaml)

The custom dataset template contains:
train: ../custom_dataset/images/train
val: ../custom_dataset/images/val
test: ../custom_dataset/images/test

is_coco: False
nc: 20

Observations

- Used for user-created datasets.
- Provides separate directories for train, validation, and test images.

- Supports custom object classes.
- Can be modified according to project requirements.

Typical YOLOv6 Dataset Structure

```
dataset/
│
├── images/
│   ├── train/
│   ├── val/
│   └── test/
│
├── labels/
│   ├── train/
│   ├── val/
│   └── test/
```

Label Format

YOLO label files use:

<class_id> <x_center> <y_center> <width> <height>

Example:

0 0.512 0.436 0.231 0.315

Where:

- 0 = Class ID
- 0.512 = X center
- 0.436 = Y center
- 0.231 = Width
- 0.315 = Height

All coordinates are normalized between 0 and 1.

Conclusion

The YOLOv6 YAML files provide essential metadata describing dataset locations, class information, and dataset type. The coco.yaml file supports 80 classes, while voc.yaml and dataset.yaml support 20 classes. These configuration files ensure that YOLOv6 correctly loads images, labels, and annotations during training and validation.

References

- YOLOv6 Official Repository
- YOLOv6 Documentation
- Ultralytics Dataset Format Guide